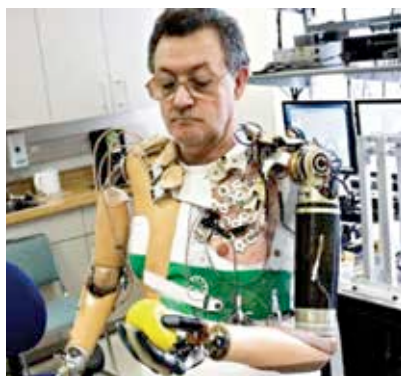




Jesse Sullivan squeezing a soft ball and using his bionic arms

7000 volts of electricity! Although it was a miracle he survived that, he did end up losing both his arms because the nerves were just too damaged. He was then operated on by doctors from Chicago, most notably Dr Todd Kuiken, who surgically redirected four nerves from his shoulder that would have controlled his arm, and placed them just under the skin on the left side of his chest. This is where they planned to attach the bionic arm, and let it read the electrical signals sent out by the nerves just under his skin. Truly a cyborg, today Jesse can move his prosthetic arm about, bend it at the elbow, open and close his hand, and not just that, but even “feel”. Not only can he adjust his grip to hold something tightly or gently, he can even feel hot or cold (presumably without pain). Because his hand nerves are under the skin of his chest, they also communicate with the nerves in his skin there. As a result, the prosthetic / bionic arm can send feedback to his chest. The tighter he squeezes his hand, the more pressure he feels on the skin on his chest, for example. Using pressure and electrical signals, Jesse can feel like the prosthetic is actually his own arm, just placed a little weirdly on his chest instead of connecting at the shoulder... sort of like having a deformation, rather than missing a whole arm.

James Young

In the world of cyborgs, James Young is kind of a super star. Yes he has a titanium prosthetic arm, but he's also a creative filmmaker and a very good communicator. He'd lost an arm and a leg in a rail accident in London in

2012. He works with Konami, yes the Japanese gaming giant, or who I like to think of as the geniuses behind *Contra* – my second favourite game of all time (everyone should know my favourite by now). Of course that's not the reason why Konami were involved. They also have this other game that some gamers might have heard of... called *Metal Gear Solid* (that was supposed to be a joke, because every gamer knows MGS). It has a character called Snake who has one bionic arm, and the idea was to find a gamer in the UK, who had an amputated arm, loved MGS, and would be willing to be fitted with a Snake-like arm. When James applied for the position, he didn't think he'd have much competition, but later found out 60 others had applied as well! James also works as a show presenter for the BBC on all things cyborg or cybernetics.



James Young with his Metal Gear Solid inspired arm

Neil Harbisson

While many may claim to be the first cyborg, Neil actually has documentary evidence. He's the first person to be legally recognised by a government as being a cyborg. He's an activist, and fights for trans-species rights. He formed a Cyborg Foundation in 2010 that raises awareness and helps people who want to be cyborgs, and also those who do not identify as humans, but as a different species. While all that is a little too weird for us, and we don't want to get into the whole debate on whether or not such identification is a mental illness or a valid identity... Instead we want to focus on the fact that he actually has an antenna attached to his skull! Harbisson was

born with a rare genetic mutation that allows him to only see the world in black and white. Instead of letting that sound depressing, he decided to let it sound colourful... literally! Yes, Neil Harbisson uses the antenna on the



Neil Harbisson listens to colours

top of his head to translate colour into sound, which is then transmitted into his skull as vibrations. While you and I merely see a colour like red, he feels it, and identifies it by its frequency! But that's not all. Neil has given access to his antenna to five trusted friends from across the globe, who can send content (and thus, the vibrations of the colours), over the internet. Not just images or colours, they can also send videos, and even sounds to him, and they can do so without any restrictions. Thus, if he's asleep, the data they send can influence his dreams... someone who is truly living his art.

He's even done some publicity stunts, by giving volunteers paint, and allowing them to paint stripes onto a canvas far away, and then receive those via the internet to his antenna and then paint a replica painting at a different and distant location. He got the painting right.

Moon Ribas

She is a Spanish performance artist and choreographer who has implanted a seismic sensor into her elbow. She literally feels the vibrations of the earth, and then incorporates that into her dance. Along with Neil Harbisson, she co-founded the Cyborg Foundation as well as the Transpecies Society. She and Harbisson both have a Bluetooth-enabled tooth to be able to accomplish what they call Transcendental Commu-

Source: thememo.com